Introduction to Software Engineering

Software Requirements Specification



Software Engineering Department
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Software Requirements Specification

Objectives

This document focus on the following topics:

- ✔ Completing the Software Requirements Specification (SRS) document with the following sections:
  - Problem Statement
  - Stakeholders
  - Functional, non-functional and domain requirements
  - Use case model and use case specifications
  - Prototype / mockups
  - Traceability and requirements review
- ✔ Including an AI usage log for this phase
- ✔ Understanding the Software Requirements Specification (SRS) document.

1 Member Contribution Assessment

ID	Name	Role(s)	Contribution (%)	Evidence
24127326	Đoàn Quốc Bảo	Team lead / PM	20	Task board: link
24127025	Lê Minh Đức	Business analyst	20	Task board: link
24127003	Vũ Trần Minh Hiếu	Designer	20	Task board: link
24127173	Nguyễn Tuấn Hùng	Developer	20	Task board: link
24127008	Trương Nhật Phát	Tester	20	Task board: link

2 Introduction

2.1 Purpose and Scope

Purpose

This document describes the functional and non-functional requirements of Portal67, a student portal designed for HCMUS. It provides a common reference for the development team and stakeholders to understand the system's features, expected behavior, and constraints.

Product Scope

Portal67 aims to provide a centralized platform for students, lecturers, teaching assistants, and administrative staff to manage academic and administrative activities.

The system includes the following main functions:

- **Student services:** Course registration and dropping, weekly timetable viewing, tuition information, academic results, GPA calculation, and administrative request submission and tracking.
- **Teaching services:** Managing component grades for assigned sections and submitting grade sheets for approval.
- **Administrative services:** Managing courses, sections, registration periods, administrative requests, tuition records, grade approvals, and announcements.
- **Common services:** User authentication, student dashboards, and in-app notifications.

The system focuses on managing academic information and administrative workflows. It does not directly process actual financial transactions or issue legally valid documents.

Intended Readers

This document is intended for:

- Project team members responsible for system analysis, design, development, and testing.
- HCMUS students who use the portal.
- Lecturers and teaching assistants who manage course activities and grades.
- Administrative staff responsible for academic and administrative operations.
- Project supervisors and other stakeholders who review the system requirements.

2.2 Glossary

Term	Definition
Portal67	The student portal system being developed for HCMUS.
Student	A user who uses the portal to register for courses, view academic information, and submit administrative requests.
Teaching Staff	Lecturers and teaching assistants responsible for teaching activities and entering student grades.
Academic Affairs	The administrative department responsible for academic operations, including registration periods and grade approval.
Course	An academic subject offered by the university in a particular academic term.
Section	A specific class instance of a course, with its own capacity, schedule, instructor assignments, and enrolled students.
Theory-Practice Sessions	The theory and practical/laboratory sessions that may belong to the same course section.
GPA	Grade Point Average, calculated from course grades and credits.

Credit	A unit representing the academic workload of a course, used in GPA and accumulated-credit calculations.
Grade Sheet	A record containing students' assessment components and scores for a particular section.
Administrative Request	A request submitted by a student for services such as verification, transcript, military postponement, or tuition extension.
Registration Period	A designated time window during which students are allowed to register for or drop courses.
Audit History	A record of actions performed on a system record, including the actor and time of the action.
In-app Notification	An alert displayed inside the portal to inform users about important events or status changes.
What-if GPA	An estimated GPA calculated using hypothetical scores without modifying official academic results.

3 Problem Statement

Business context:

HCMUS provides numerous academic and administrative services to students, lecturers, and staff. These services include course registration, academic result management, tuition management, document requests, and other activities.

Students regularly interact with these services throughout their university years. Lecturers are responsible for academic tasks such as entering grades, while administrative staff handle tasks such as managing tuition records and processing student requests.

However, these activities are currently handled through different university and faculty platforms. This creates a need for a more centralized and convenient way to access and manage common academic and administrative tasks.

Portal67 is proposed as a web application that brings these functions together in one place. It aims to provide a consistent interface for students, lecturers, and administrative staff, making their daily tasks easier to access and manage.

Current problems:

The main problem is that university services are spread across multiple websites and platforms. Students have to visit different websites depending on the task they want to complete.

For instance:

+ To check for the status of having participated in specialized topics (“Chuyên đề A”, “Chuyên đề B”), students need to access this website: https://bit.ly/tracuu-ctsv-vnuhcmus;
+ To pay for tuition fees, they must do it in: https://hocphi.hcmus.edu.vn/dang-nhap;
+ To know their conduct score (“Điểm rèn luyện”), follow this link: https://hcmus.drl.vn/;

+ To register for courses and check grades, they must look up the faculty's portal;
+ To sign up for a specialized topic ("Chuyên đề A"), they need to visit the university's portal;
+ And they have to provide feedback on the student's portal every semester.

Having to remember and switch between many different platforms is inconvenient and makes it painfully harder for students to manage their university life efficiently.

Additionally, there are a few problems of the current system that need addressing:

1. Course registration and seat availability

During registration periods, many students access the registration portal simultaneously. This can cause slow response times and make the system difficult to use.

The first-come, first-served registration process also creates challenges when many students compete for a limited number of seats. The system must handle concurrent registration requests correctly to prevent multiple students from claiming the same remaining seat or exceeding a section's capacity.

2. Timetable planning

The faculty's portal mainly displays available classes as a list sorted by course name and class ID. Class schedules are not presented in a convenient way, making it difficult for students to compare different class options.

Students who want to arrange their courses into a timetable with fewer days at university or avoid overlapping classes must often compare schedules manually. This requires considerable effort and trial and error.

3. Academic result management

Students need to monitor their academic results, accumulated credits, and semester or cumulative GPA throughout their studies. However, understanding their academic progress may require checking multiple records or calculating results manually.

Lecturers also need a structured way to enter and manage component grades. Grade submission, review, approval, and publication involve multiple steps and different responsibilities. Without a clear workflow, it can be difficult to track the status of grade sheets and ensure that only authorized staff can finalize or modify grades.

Students may also need to request a grade review or correction when they believe that a published result is incorrect.

4. Tuition management

Students need to keep track of tuition charges, payment records, outstanding balances, and payment status.

When tuition information is maintained separately from other academic information, students may need to access another platform to understand their financial obligations. Administrative staff also need a convenient way to look up student records and update payment information.

5. Administrative document requests

Throughout their university years, students may need to submit various administrative requests, such as student status confirmations, transcripts, military service postponement documents, and tuition extension requests.

However, students may not have a convenient way to track the progress of these requests. After submitting a request, they may not know whether it has been received, is being processed, requires additional information, or has been approved or rejected.

This lack of information makes it difficult for students to know when they should follow up or take further action. Administrative staff also need a convenient way to review requests, provide feedback, and record their decisions.

6. Lack of centralized information and communication

Students need to keep track of several types of information, including daily schedules, academic results, GPA, earned credits, pending administrative requests, and announcements.

Since these functions are handled by different systems, students may have to visit multiple websites to obtain a complete overview of their academic situation.

Furthermore, important updates such as grade publication, registration period changes, and administrative request decisions need to reach the relevant users. Without a centralized notification and announcement mechanism, users may miss important information or need to check different platforms regularly.

These problems create a need for a centralized system that simplifies common university activities and provides a more consistent experience for students, lecturers, and administrative staff.

Goals

The main goal of Portal67 is to provide a centralized web application for common academic and administrative activities at HCMUS.

To address the problems described above, the project aims to achieve the following goals:

G01 – Centralize University Services: Provide a single interface where users can access common academic and administrative functions, reducing the need to switch between different platforms.

G02 – Improve Course Registration and Management: Allow students to browse available courses and specialized topics, check section information and seat availability, and register for or drop courses. The system should validate registration eligibility, registration periods, and section capacity while handling concurrent registration requests correctly. Administrative staff should also be able to configure courses, sections, and registration periods.

G03 – Improve Timetable Planning: Display students' weekly timetables and provide tools for planning alternative schedules. The system should help students identify overlapping sessions and compare available course combinations. It should also provide access to relevant examination schedule information.

G04 – Simplify Tuition Management: Allow students to view semester tuition charges, payment records, outstanding balances, and payment status. Administrative staff should be able to update simulated payment records.

G05 – Improve Academic Result Management: Allow lecturers and teaching assistants to enter component grades for assigned courses. Support grade sheet submission, review, approval, and finalization by authorized staff. Students should be able to view finalized results, accumulated credits, and estimated semester and cumulative GPAs based on predefined grading rules.

G06 – Improve Administrative Request Management: Allow students to submit supported administrative requests, attach required documents, and track their progress. Administrative staff should be able to review requests, approve or reject them, request additional information, and record their completion.

G07 – Improve User Access and Communication: Provide authentication and role-based permissions for students, teaching staff, and administrative staff. The system should also support user profiles, targeted announcements, and notifications for important academic and administrative events.

G08 – Support Additional Academic Workflows: Provide supporting functions such as cross-registration petitions, specialized topic preference submission, grade review requests, approved grade corrections, and personal timetable export.

These goals will be evaluated through a working prototype that demonstrates the main functions and provides a convenient user experience.

Operating Environment

Portal67 will be developed as a web application using a client-server architecture.

The system will be accessible through modern web browsers on desktop, laptop, and mobile devices. Users will interact with the frontend, which communicates with the backend through REST APIs. The backend will handle business logic, validate requests, enforce access permissions, and communicate with the database.

The system will support the following user roles:

- Students: Browse available courses, register for or drop sections, plan timetables, view academic results and tuition information, submit administrative requests, and access announcements and notifications.
- Lecturers: Manage academic results for assigned courses, enter component grades, and submit grade sheets for approval.
- Teaching Assistants: Assist lecturers with assigned academic tasks, such as entering component grades, according to their permissions.
- Primary Lecturers: Manage and submit grade sheets for their assigned sections.
- Administrative Staff: Manage courses and sections, update tuition records, review administrative requests, and publish announcements.

- Academic Affairs Staff: Manage registration periods and review, approve, and finalize academic results.

The system will also distinguish students based on relevant academic information, such as their study track, cohort, or eligibility for particular registration activities.

The database will store information such as user accounts, roles, student profiles, courses, sections, class schedules, enrollments, academic results, tuition records, administrative requests, announcements, and notifications.

Design and Implementation Constraints

The project plans to use the following technologies:

Component	Planned Technology	Purpose
Frontend	Svelte, Tailwind CSS	Build a responsive and modular user interface.
Backend	NestJS or Go	Implement business logic and REST APIs.
Database	PostgreSQL	Store and manage relational data.
Caching and concurrency	Redis	Support caching, session management, and concurrent registration operations.

Deployment	Docker, Nginx, Kubernetes	Support containerization, reverse proxying, and deployment.
Testing and CI	Jest, GitHub Actions	Perform automated testing and manage CI workflows.

The final technology choices may depend on the team's experience and the available development time. The implementation should also maintain consistency between the selected technologies, system architecture, and testing tools.

The project has a limited development period, with implementation scheduled for Weeks 8–10. Therefore, the team must prioritize requirements and organize implementation work carefully.

The following constraints apply to the prototype:

- The system will use sample or seeded data and will not synchronize academic, tuition, or administrative records with official HCMUS systems.
- Authentication will use provisioned accounts for the prototype. Institutional OAuth/SSO may be supported as an alternative where the required identity provider is available.
- Course registration will support eligibility checks, registration periods, seat limits, duplicate registration prevention, and timetable conflict checking. It will also handle concurrent registration requests to prevent over-allocation. However, it will not implement a complete university registration process or guarantee protection against every form of automated registration abuse.
- Tuition management will use simulated payment records. The system will not process real financial transactions or connect to banking services.
- GPA calculations will follow predefined or configurable sample rules and may not cover every official university grading policy.

- Document requests will support submission, review, approval, rejection, clarification, and status tracking, but the system will not issue legally valid documents.
- Academic results and grade workflows will use sample academic data. Grade approval, publication, review, and correction will be demonstrated within the prototype.
- Security requirements will include authentication, role-based authorization, password protection, token management, and protection against unauthorized access to other users' data. However, the prototype will not be considered a production-certified university system.
- Performance, concurrency, reliability, and responsive interface requirements will be verified under the testing conditions defined in the SRS. These results do not guarantee production-level performance under every real-world workload.
- Advanced deployment and scaling features, such as Kubernetes and multi-service deployment, will depend on the final architecture and available time.

Expected Outcome

The expected outcome is a working prototype of Portal67 that brings common academic and administrative functions into a single web application.

The prototype should demonstrate the main workflows for students, lecturers, teaching assistants, and administrative staff. These include course registration, timetable planning, tuition management, academic result management, administrative requests, user access control, announcements, and notifications.

The system will also demonstrate important supporting requirements, such as handling concurrent registration requests, protecting user information, validating submitted data, and maintaining consistent records throughout academic and administrative workflows.

Its success will be evaluated based on the correctness of its functional requirements, compliance with the specified non-functional requirements, ease of use, and fulfillment of the course project objectives.

Portal67 is developed as a course project prototype. It is not intended to replace the official HCMUS Portal or other existing university systems.

4 Changes since PA1

Changes Since PA1

This section records the changes and refinements made to the project requirements between PA1 and PA2. Since PA1 has already been submitted, its contents remain the original project baseline. The requirements in PA2 provide a more detailed description of the system based on further analysis of user needs, system workflows, and technical constraints.

The overall project goal remains unchanged: to develop a centralized academic and administrative portal for HCMUS students and staff. However, several features have been expanded, supporting capabilities have been identified, and some implementation details have been clarified.

1. Refinements to Existing PA1 Features

PA1 Feature	Changes and Refinements in PA2
F01 – Course and Specialized Topic Registration	Expanded into a more detailed registration workflow. Students can browse, search, and filter eligible course sections, register for and drop sections, and receive timetable conflict warnings. The requirements also clarify section capacity enforcement, registration periods, program eligibility, and integrated Theory-Practice enrollment. Cross-registration petitions and trial timetable planning are also described as supporting capabilities.

F02 – Tuition Management	Clarified how tuition obligations are calculated based on enrolled credits and configured tuition rates. Administrative staff can update simulated payment records and statuses. The prototype does not process real payments or connect to an external payment gateway.
F03 – Academic Result Management	Expanded to cover the complete grade management workflow. Assigned lecturers and teaching assistants can enter scores, Primary Lecturers can submit grade sheets, and Academic Affairs staff can review, approve, and publish finalized grades. GPA and earned credits are recalculated after publication. A separate grade-correction workflow is also introduced for finalized grades.
F04 – Course and Exam Scheduling	Expanded into a personalized weekly timetable showing enrolled Theory and Practice sessions, room information, campus, and instructors. Examination schedules are also included when examination data is available. Schedule export is identified as an additional convenience capability.
F05 – Resource Management	Resource management remains part of the original PA1 proposal. However, a complete set of functional requirements and acceptance criteria for room and equipment availability and booking has not yet been defined in the current PA2 requirements. Therefore, it is not treated as a fully specified capability in this version.

F06 – Document Request and Approval	Expanded to describe the full administrative request workflow, including online submission, supporting file attachments, staff review, approval, rejection, requests for additional information, and status history. The prototype tracks requests but does not issue legally valid official documents.
F07 – Role-Based Access Control (RBAC)	Elaborated through authentication, role-based authorization, and record-level access restrictions. The requirements now describe how students, lecturers, teaching assistants, and administrative staff access only the information and operations permitted by their roles.
F08 – Basic Validation	Expanded into more specific validation rules across the system, including registration eligibility, duplicate enrollment prevention, section capacity, timetable conflicts, grade score validation, assessment weight validation, and administrative workflow restrictions.

2. Additional Capabilities Identified in PA2

Further analysis identified several supporting capabilities that were not explicitly described as separate features in PA1. These capabilities improve the completeness and usability of the portal.

Additional Capability	Description

Authentication and Account Management	Defines how pre-provisioned students and staff authenticate using institutional identifiers and passwords. Optional institutional OAuth/SSO support is considered if an identity provider is available. Demo accounts are also provided for project assessment and testing.
Student Dashboard and Profile	Provides a centralized overview of academic information, including daily schedules, GPA, earned credits, pending requests, and announcements. Students can update permitted contact details while academic information remains protected.
Schedule Planning	Allows students to create and inspect trial timetable combinations without registering for courses or reserving seats.
Specialization Preferences	Allows eligible students to submit ranked preferences for academic specialization tracks during designated periods.
Cross-Registration Petitions	Allows eligible Project-Based students to request enrollment in permitted Standard-track sections, subject to the required approval process.

Grade Review and Correction	Supports student grade-review requests during configured appeal periods and provides an auditable workflow for correcting finalized grades.
Academic Administration	Provides authorized staff with functions for configuring registration periods, managing courses and sections, maintaining Theory-Practice schedules, and assigning lecturers and teaching assistants.
In-App Notifications	Provides notifications for important events, such as grade publication, administrative request status changes, and registration-period openings.
Announcements	Allows authorized staff to publish announcements for all users or specific audiences, such as a program track or faculty.

These capabilities are assigned individual priorities in the PA2 Functional Requirements. Their inclusion does not mean that every capability has the same implementation priority as the original Must features.

3. Non-Functional and Technical Requirement Refinements

PA1 established the general technical direction of the project. PA2 further defines quality expectations and system constraints so that they can be evaluated during implementation and testing.

The main refinements include:

- **Performance:** Introduces measurable response-time and load-testing targets for important endpoints and registration operations.
- **Concurrency and Data Integrity:** Defines how the system should prevent section over-allocation and maintain consistent enrollment records when multiple students register simultaneously.
- **Security:** Specifies authentication, token expiration, password protection, role-based authorization, and protection against unauthorized access to other users' records.
- **Usability and Accessibility:** Defines responsive layout expectations, keyboard accessibility, visible focus indicators, and text contrast requirements.
- **Reliability:** Requires important operations, such as enrollment, grade publication, and request updates, to avoid partially updated records when an operation fails.
- **Maintainability and Portability:** Introduces automated testing expectations, code coverage targets, and reproducible development environment requirements.
- **System Availability:** Adds expectations for health checks and for registration activity not to disrupt unrelated portal functions.

Implementation-specific decisions, such as the use of Redis atomic operations, background workers, or a particular backend framework, are treated as design and architecture decisions rather than user-facing functional requirements.

4. Priority and Scope Clarifications

The PA2 requirements provide more detailed priorities for individual system capabilities. The original PA1 priorities remain the reference point, while any differences are recorded as refinements rather than changes to the already-submitted document.

In particular:

- Tuition management remains a **Must** feature, consistent with PA1. Its implementation is limited to estimated tuition obligations and simulated payment records.

- Resource management remains in the original PA1 proposal, but its detailed requirements have not yet been completed in the current PA2 specification.
- Supporting capabilities such as schedule planning, specialization preferences, cross-registration, grade review, and schedule export have their own priorities and are not automatically treated as core Must features.
- Authentication, dashboard functionality, announcements, notifications, and academic administration are described in greater detail in PA2 to support the operation of the overall portal.

5. Prototype Assumptions and Limitations

The requirements have also been clarified to distinguish the intended prototype from a production university information system.

The following limitations apply:

- The system uses prepared sample data and pre-provisioned accounts rather than integrating with the official HCMUS student database.
- Institutional OAuth/SSO is conditional on the availability of a suitable identity provider. Local institutional-ID and password authentication is the baseline prototype approach.
- Tuition payment records are simulated; no real financial transactions are processed.
- Administrative document requests can be submitted and tracked, but the system does not issue legally valid official documents.
- Academic policies, tuition rates, grading conversions, GPA calculations, and other institutional rules use configurable prototype assumptions unless their official sources have been verified.
- The system is developed as a course-project prototype and does not claim production-level security certification or availability.

Summary

PA2 does not replace the original project direction established in PA1. Instead, it elaborates the original features into more specific and testable requirements, identifies supporting capabilities, and defines clearer quality expectations and prototype limitations.

These refinements provide a more complete basis for the subsequent design, implementation, and testing phases while preserving the original project objectives

5 Stakeholders

#	Stakeholder	Description	Interests / concerns
1	Project Sponsor	Individuals or teams providing resources, support, or direction for the project.	They want the project to meet its objectives, demonstrate practical value, and be completed within the planned schedule and resources.
2	Students	HCMUS students, including Standard and Project-Based students, who use the portal for academic and administrative activities.	They want convenient course registration, accurate academic records, transparent grading, easy access to tuition information, and timely updates on administrative requests.
3	Lecturers and Teaching Assistants	Teaching staff assigned to course sections. They are responsible for entering and updating component grades. The designated Primary Lecturer also submits completed grade sheets to Academic Affairs.	They want efficient grade management, clear permission boundaries, accurate grade calculations, and a reliable submission process.
4	Academic Affairs Staff (PĐT / VP Đề án)	Authorized staff responsible for academic operations, including course and section management, registration periods, grade approval and finalization, and grade appeal periods.	They want accurate academic records, reliable administrative tools, proper approval workflows, and control over academic operations.
5	Student Affairs Staff (Phòng CTSV)	Administrative officers responsible for reviewing and processing student requests, including document requests	They want an organized request queue, sufficient information for reviewing requests, clear status

		and related administrative procedures.	tracking, and reliable communication with students.
6	Development Team	Team members responsible for analyzing, designing, implementing, and testing the Portal67 prototype.	They want clear requirements, manageable tasks, suitable technical solutions, and a system that meets the project objectives.
7	Project Advisor / Instructor	The instructor responsible for supervising and evaluating the project and its development process.	They want proper requirements analysis, system design, implementation, testing, teamwork, and compliance with course requirements.
8	Institutional Identity Provider	The university's external authentication service that supports institutional OAuth/SSO login.	It requires secure authentication integration, proper identity verification, and correct mapping of institutional accounts to Portal67 users.

6 Requirements

6.1 Functional Requirements

ID	Requirement	Feature	Priority	Source
FR-01	The system shall authenticate users via institutional identifier (Student ID MSSV / Staff ID MSCB) and password credentials or via institutional OAuth/SSO, returning cryptographically signed JWT tokens (Access Token lifetime <= 15 min, Refresh Token lifetime <= 7 days). Public self-registration is strictly omitted; all accounts are pre-provisioned by administration.	F01	Must	HCMUS Identity Management Standards & PA1 Project Baseline
FR-02	The system shall provide pre-configured demo switchers representing Student (Standard), Student (Project-Based CLC), Lecturer, and Academic Affairs Staff for course evaluation and testing.	F01	Should	NMCNPM Course Project Assessment Guidelines
FR-03	The system shall enforce backend role-based access control (RBAC) on all protected API routes, rejecting unprivileged or cross-tenant data requests with HTTP 403 Forbidden.	F01	Must	ISO/IEC 27001 Access Control Policy & Solution Architecture v3.0
FR-04	The system shall aggregate and render a real-time student dashboard displaying daily class schedules, cumulative GPA, total earned credits, pending document requests, and active announcements.	F02	Must	Stakeholder Interview: FIT Undergraduate Students
FR-05	The system shall display the student's biographical data, academic program track (STANDARD vs. PROJECT_BASED), cohort, and advisor, while allowing students to update personal contact details.	F02	Should	HCMUS Student Handbook 2026 (Sổ tay sinh viên)
FR-06	The system shall allow students to browse, search, and filter course offerings with smart program-cohort catalog partitioning (STANDARD vs PROJECT_BASED), department, campus location (CS1 vs CS2), and integrated Theory-Practice session schedules.	F03	Must	FIT Course Offering & Registration Regulations & ADR-008
FR-07	The system shall allow students to enroll in open course sections during active registration windows; for courses with lab components (has_lab = TRUE), enrolling in a section automatically registers and	F03	Must	FIT Credit-Based Training

	schedules both the Theory and Practice sessions of that section, atomically reserving the section seat via high-concurrency in-memory engine.			Guidelines & ADR-007
FR-08	The system shall allow students to drop registered sections within permitted drop windows, atomically relinquishing the allocated section seat (releasing both Theory and Practice scheduled slots) in memory and updating relational records.	F03	Must	HCMUS Academic Affairs Drop Regulations
FR-09	The system shall detect overlapping time slots between newly selected and enrolled sections, displaying prominent non-blocking amber advisory warnings (CONFLICT_DETECTED) without preventing submission.	F03	Must	HCMUS Academic Advisory Policy & ADR-004
FR-10	The system shall provide an interactive schedule planner allowing students to generate and inspect trial timetable combinations without committing to official database enrollment or claiming seats.	F03	Should	Undergraduate Student Council Survey & Solution Architecture
FR-11	The system shall enable qualified students completing Year 2 to submit ranked preferences for academic specialization tracks during designated application periods.	F03	Could	FIT Academic Handbook — Specialization Allocation Procedures
FR-12	The system shall allow Project-Based students to submit cross-registration petitions for eligible Standard track sections (is_cross_enrollment_allowed = TRUE) with advisory tuition settlement warnings.	F03	Should	VP Đề án (CTĐA) Cross-Enrollment Policy & ADR-008
FR-13	The system shall render an interactive weekly timetable of enrolled sections filtered by semester, week number, and study day, indicating room code, building, campus (CS1/CS2), section code, and instructor.	F04	Must	HCMUS Timetable Standards & ADR-008
FR-14	The system shall export the user's personalized weekly schedule into standard iCalendar (.ics) format or printable PDF view.	F04	Could	Usability Improvement Proposal
FR-15	The system shall render student semester and cumulative grade reports (10-point scale, letter grades, 4-point GPA, accumulated credits) broken down by dynamic component rubrics for finalized academic terms.	F05	Must	MoET Circular 08/2021/TT-BGDĐT on Undergraduate Grading
FR-16	The system shall provide an interactive "What-If" GPA projection tool enabling students to simulate hypothetical target scores for current terms and evaluate impact on cumulative GPA.	F05	Should	FIT Academic Advisor Committee Recommendations

FR-17	The system shall allow assigned primary lecturers and teaching assistants to input and update continuous component scores across dynamic, configurable assessment rubrics (academic.grade_components).	F05	Must	FIT Lecturer Grading Regulations & CHG-05
FR-18	The system shall permit only the designated Primary Lecturer (is_primary = TRUE) to lock and submit completed section grade sheets to Academic Affairs, transitioning status from Draft to Submitted.	F05	Must	FIT Department Regulations on Separation of Grading Duties
FR-19	The system shall authorize Academic Affairs staff to review, approve, and officially publish submitted grade sheets (Finalized), automatically triggering GPA and credit recalculations for all enrolled students.	F05	Must	HCMUS Academic Affairs Grade Administration Policy
FR-20	The system shall enforce an immutable change-request workflow for modifying finalized grades, recording prior score, new score, justification reason, and approving officer ID in Grade_Correction_Log.	F05	Should	ISO/IEC 27001 Audit Trail & MoET Examination Inspection Protocol
FR-21	The system shall enable students to submit formal administrative certificate applications online (enrollment verification, military postponement, official transcript, tuition extension) with digital document upload.	F06	Must	HCMUS Student Affairs Office (Phòng CTSV) Regulations
FR-22	The system shall provide administrative staff with a centralized review queue to filter, inspect, approve, return for clarification, or reject student requests.	F06	Must	CTSV Officer Workflow Interview & Specifications
FR-23	The system shall enable students to track real-time progression of submitted requests through a defined unidirectional status lifecycle.	F06	Must	Student Experience Improvement Baseline
FR-24	The system shall allow students to lodge formal grade review requests (Phúc khảo) within the appeal window configured and opened by Academic Affairs staff following official grade finalization.	F06	Should	HCMUS Academic Regulations on Grade Appeals (Quy chế Phúc khảo)
FR-25	The system shall permit administrators (PĐT or VP Đề án) to configure, open, suspend, and close term course registration periods segregated by academic program track and cohort.	F07	Must	PĐT & VP Đề án Operational Workflow Manual
FR-26	The system shall allow administrative staff to provision and manage courses, section Theory-Practice session schedules, classroom	F07	Must	FIT Curriculum Administration Standards & ADR-008

	facilities with campus allocations (CS1/CS2), and co-teaching/TA assignments.			
FR-27	The system shall display semester tuition obligations calculated by enrolled credit hours and allow staff to update simulated payment statuses.	F07	Should	HCMUS Tuition Billing Regulations
FR-28	The system shall dispatch real-time asynchronous in-app notifications upon critical lifecycle events (grade finalized, request status changed, registration window opened) via decoupled background workers.	F08	Must	Solution Architecture v3.0 & ADR-007
FR-29	The system shall authorize administrative staff to broadcast global, program-specific, or faculty-wide markdown announcements visible on user dashboards.	F08	Must	FIT Communication & Public Information Guidelines

6.2 Non-Functional Requirements

ID	Category	Requirement (measurable)	How it will be verified
NFR-01	Performance	Under 1,000 sustained HTTP GET requests to /api/academic/sections and /api/grades/mine, the system shall return responses with p95 <= 300 ms and p99 <= 800 ms.	Measured using k6 load-testing scripts executing against the containerized reverse proxy gateway.
NFR-02	Performance	Under 100 concurrent virtual users executing section enrollment transactions, the registration-service shall execute Redis Lua seat reservations in <= 10 ms and complete end-to-end HTTP requests with p95 <= 300 ms and max <= 800 ms.	Measured using k6 concurrent write scenarios targeting /api/registration/sections/register.
NFR-03	Scalability	Under a 3-minute sustained stress test, the system shall sustain >= 100 RPS overall and >= 500 RPS on the in-memory seat reservation engine with an HTTP error rate < 0.1%.	Verified via automated k6 stress-test harness running in standard Docker Compose test node.
NFR-04	Concurrency / Integrity	When 50 concurrent threads simultaneously attempt to claim the final 1 remaining seat in a section, the system shall maintain exactly 0 over-allocations for the section (encompassing both integrated Theory and Practice schedules).	Automated multi-threaded integration test issuing concurrent POST requests; verified via database count assertions.
NFR-05	Security	The system shall hash 100% of stored user passwords using Argon2id (memory: 64MB,	Static source code security review and

		iterations: 3, parallelism: 2) or bcrypt (work factor >= 12).	direct cryptographic audit of auth.accounts database records.

NFR-06	Security	User access tokens shall have a maximum Time-To-Live (TTL) of <= 15 minutes, and refresh tokens shall have a maximum TTL of <= 7 days.	Automated unit test suite verifying token expiration enforcement past 901 seconds.
NFR-07	Security	The system shall prevent Insecure Direct Object References (IDOR); cross-student grade retrieval or unauthorized data access shall yield 0 data leaks and return HTTP 403 Forbidden.	Automated security integration test suite executing unauthorized cross-tenant requests.
NFR-08	Usability	The web interface shall display 0 horizontal scrollbars or broken layouts across responsive breakpoints: 375px (Mobile), 768px (Tablet), and 1280px+ (Desktop).	Visual regression testing using Playwright across configured viewport matrix.
NFR-09	Usability	Under simulated 4G mobile network conditions, the web application shall achieve First Contentful Paint (FCP) <= 1.5 seconds, with an initial production JS+CSS bundle <= 450 KB (gzipped).	Automated Google Lighthouse CI audit running against production frontend build.
NFR-10	Usability / Accessibility	The portal shall achieve WCAG 2.1 AA compliance with text color contrast >= 4.5:1, 100% keyboard operability, visible focus indicators, and an automated Lighthouse Accessibility score >= 95/100.	Automated Axe-core audit, Lighthouse Accessibility scan, and manual NVDA/ChromeVox screen reader evaluation.
NFR-11	Reliability	The system shall ensure 100% ACID compliance on grade finalization and course registration write-behind persistence queues.	Integration tests simulating transient database failures during transaction processing, asserting zero corrupted partial states.
NFR-12	Observability	The /health endpoint of each microservice shall respond with HTTP status 200 in <= 50 ms.	Continuous curl health-check polling every 10 seconds across all 5 backend microservices during staging runtime.
NFR-13	Maintainability	The core backend business domain packages shall maintain a statement test coverage of >= 70%.	Go test coverage analysis: go test -coverprofile=coverage.out ./...
NFR-14	Portability	The entire multi-service application environment shall boot from a clean git checkout to a fully	Timed clean deployment script on a standard dual-core

		reachable operational state via docker compose up -d in <= 60 seconds.	development machine with pre-cached Docker images.
NFR-15	Architecture	Flash-crowd traffic spikes saturating the registration-service (100% CPU) shall result in 0% cascade failure on grade inspection and document submission services.	Chaos engineering test: running full load against registration while simultaneously verifying HTTP 200 on /api/grades/mine.

6.3 Domain Requirements and Business Rules

ID	Rule	Source (policy, law, standard)
BR-01	**Active Registration Window Required:** Section registration, modification, and dropping are permitted strictly when the registration period assigned to the student's program track (STANDARD vs PROJECT_BASED) and cohort is in the operational status OPEN.	HCMUS Academic Affairs Policy, Decision No. 128/QĐ-KHTN on Course Registration
BR-02	**No Duplicate Course Code Enrollment:** A student cannot hold simultaneous active enrollments in multiple sections belonging to the same root course (course_id) within the same academic semester, even under different section codes.	MoET Circular 08/2021/TT-BGDĐT on Credit-Based Higher Education, Art. 5
BR-03	**Advisory Non-Blocking Timetable Overlaps:** Time slot collisions between newly selected and enrolled sections generate prominent UI warnings (CONFLICT_DETECTED) but do not programmatically block registration submission.	HCMUS Academic Advisor Guidance Policy & ADR-004
BR-04	**Hard Section Capacity Enforcement:** Total enrolled students in any section must never exceed section.max_capacity. Any concurrent registration attempting to exceed capacity must be rejected immediately.	FIT Classroom & Laboratory Safety Capacity Regulations
BR-05	**Decoupled Schedule Planning:** Saved schedule plans constructed in the Schedule Planner represent draft advisory combinations; they confer no reserved seat rights or official enrollment claims until officially submitted and committed.	Portal67 Solution Architecture v3.0 Specification
BR-06	**Program Track Cohort Isolation:** Standard track students (STANDARD) cannot register Project-Based sections (PROJECT_BASED). Project-Based students may only enroll in Standard sections when is_cross_enrollment_allowed = TRUE under formal academic petition.	VP Đề án (CTĐA) Training Scheme Regulations & ADR-008
BR-07	**Integrated Theory-Practice Enrollment:** For courses with laboratory components (has_lab = TRUE), each course section encapsulates both Theory and Practice lecture/lab schedules. Registering for the section automatically enrolls and schedules both Theory and Practice sessions for that single section	FIT Laboratory Training & Course Syllabus Policy & ADR-008

	without separate or unbundled registration. Selecting a section commits the student to both required schedules.	
BR-08	**In-Memory Atomic Seat Reservation:** During peak registration periods, seat quotas are reserved atomically via Redis Lua scripts (seat_reserve.lua) prior to asynchronous relational persistence, guaranteeing 0 over-allocations without database row lock contention.	High-Concurrency System Design Standards & ADR-007
BR-09	**Standard 10-Point Component Grading Scale:** All continuous and final grade components must be evaluated on a scale of 0.0 to 10.0 with a precision of one decimal place.	MoET Circular 08/2021/TT-BGDĐT, Art. 9 (Academic Evaluation)
BR-10	**Dynamic Assessment Rubric Weighting:** A course's grade composition is defined dynamically via k grade components (academic.grade_components). The sum of all active component weights must equal exactly 100% (1.00) prior to grade sheet submission.	FIT Academic Evaluation Framework & CHG-05 (BCNF Normalization)
BR-11	**Semester and Cumulative GPA Computation:** Semester GPA is calculated as the credit-weighted average of course grades to two decimal places. Cumulative GPA considers only the highest earned grade for retaken courses according to faculty credit regulations.	HCMUS Undergraduate Academic Regulations & Credit Training Policy
BR-12	**Separation of Grading Duties (Four-Eyes Principle):** Lecturers hold authority only to input and submit section grade sheets (Draft -> Submitted). Only authorized Academic Affairs personnel (ADM) hold authority to approve, publish, and finalize official grades (Submitted -> Finalized).	FIT Department Grading Governance Protocol & Four-Eyes Principle
BR-13	**Immutability of Finalized Grades & Audit Trail:** A finalized grade sheet cannot be modified via standard update APIs. Any post-finalization grade amendment requires a formal rectification record (Grade_Correction_Log) specifying prior score, new score, justification reason, and approving officer ID.	ISO/IEC 27001 Audit Trail Standards & MoET Inspection Protocol
BR-14	**Multi-Lecturer Assignment Permissions:** In co-taught sections (academic.section_lecturers), co-professors and TAs may enter and update continuous component scores; however, only the designated Primary Lecturer (is_primary = TRUE) holds permission to lock and submit the completed grade sheet.	FIT Faculty Teaching Assignment Regulations
BR-15	**Immutable Request State Machine:** Administrative requests adhere to a unidirectional progression: Submitted -> Under_Review -> Approved / Rejected -> Completed. Requests cannot be deleted from the audit ledger once submitted.	HCMUS Student Affairs (Phòng CTSV) Administrative Procedure Standard
BR-16	**Information Revision Cycle:** If an administrative officer marks a request as Requires_Information, the request transitions back to a student-editable state while preserving all historical audit notes and timestamps.	Phòng CTSV Administrative Petition Protocol
BR-17	**Configurable Grade Appeal Window (Phúc khảo):** Grade appeal applications are accepted strictly within the review/appeal window configured and opened by Academic Affairs staff (appeal_start_date to appeal_end_date) following	HCMUS Student Handbook & Examination Regulations (Quy chế Khảo thí)

	official grade finalization. Academic Affairs retains full authority to set and adjust the appeal duration per term or department.	
BR-18	**Immutable Student Identifier:** The student ID (MSSV) serves as the persistent, immutable institutional key within the academic domain. An account's email may be updated, but MSSV cannot be changed.	VNU-HCM Student Identification Standards
BR-19	**Protected Academic Profile Fields:** Students are forbidden from directly modifying their core academic attributes (MSSV, enrolled program track, cohort, admitted major, cumulative credits, or academic status).	HCMUS Academic Affairs Database Integrity Policy
BR-20	**Pre-Provisioned Accounts & No Public Self-Registration:** As an institutional university portal, Portal67 strictly prohibits public self-registration. All student, lecturer, and administrative staff accounts are pre-provisioned by university administration or synchronized via institutional directories. Authentication requires institutional identifier (MSSV / MSCB) and password, or authorized institutional OAuth/SSO.	HCMUS Information Security & User Identity Administration Policy

6.4 Acceptance Criteria

Feature F01: User Authentication & Access Control (FR-01, FR-03)

Scenario AC-F01-01: Successful Authentication and Scoped JWT Token Issuance (Happy Path)

Scenario: Authenticated user receives valid scoped access and refresh tokens

- **Given** A pre-provisioned user exists with institutional identifier "24127003" (MSSV) and password "CorrectPassword123!"
- **And** The account status is "ACTIVE" with role "STUDENT"
- **When** The user submits a POST request to "/api/auth/login" with identifier "24127003" and password
- **Then** The system returns HTTP status 200 OK
- **And** The response body contains an Access Token with expiry <= 15 minutes
- **And** The response body contains a Refresh Token with expiry <= 7 days
- **And** The Access Token claims contain "sub", "role: STUDENT", and student MSSV "24127003"

Scenario AC-F01-02: Backend Authorization Guard Rejects Unauthorized Access (Edge Case)

Scenario: Student attempts to access administrative endpoint

- **Given** A student is authenticated with role "STUDENT" and holds a valid Bearer token
- **When** The student sends a GET request to "/api/affairs/requests/queue"
- **Then** The API gateway rejects the request with HTTP status 403 Forbidden

- **And** Returns an error payload: '{"code": "FORBIDDEN", "message": "Insufficient permissions"}'
- **And** An audit security log entry is recorded with actor ID, IP address, and timestamp

Scenario AC-F01-03: Successful Authentication via Institutional OAuth/SSO

Scenario: Pre-provisioned user authenticates via Institutional OAuth/SSO

- **Given** An institutional user account "24127003" is pre-provisioned in Portal67 with role "STUDENT"
- **And** The user authenticates successfully via university OAuth identity provider with email "24127003@student.hcmus.edu.vn"
- **When** The OAuth callback returns authorization code to "/api/auth/oauth/callback"
- **Then** The system verifies the token, resolves identity to student MSSV "24127003"
- **And** Issues an Access Token (expiry <= 15 minutes) and Refresh Token (expiry <= 7 days)

Feature F02: Student Overview & Academic Profile (FR-04)

Scenario AC-F02-01: Student Dashboard Overview Rendered with Consolidated Academic Data

Scenario: Student views dashboard upon successful login

- **Given** Student "24127003" is authenticated and navigates to "/student/dashboard"
- **And** The student has 3 enrolled classes for the current semester
- **And** The student has 1 pending document request "REQ-2026-001"
- **When** The dashboard page completes loading
- **Then** The interface displays today's scheduled class sessions with room codes and times
- **And** Displays the student's cumulative GPA (e.g., "8.75 / 10.0") and total credits earned (e.g., "68")
- **And** Displays the active document request status badge as "Under_Review"
- **And** Loads recent faculty markdown announcements in the notification pane

Feature F03: Course & Section Registration (FR-06, FR-07, FR-08, FR-09)

Scenario AC-F03-01: Successful Section Enrollment with Integrated Theory and Practice Sessions (Happy Path)

Scenario: Student successfully enrolls in a course section containing Theory and Practice schedules

- **Given** The student "24127003" is authenticated and on the Course Registration page
- **And** The registration window for semester "2026-1" is currently "OPEN"
- **And** Course "CSC10002" (Introduction to Programming) has has_lab = TRUE

- **And** Section "CSC10002_01" encapsulates both Theory (Monday 07:30 - 10:00) and Practice (Wednesday 13:00 - 16:30)
- **And** Section "CSC10002_01" has 38 enrolled out of 40 max capacity
- **And** The student is not currently enrolled in any section of course "CSC10002"
- **When** The student selects section "CSC10002_01"
- **And** Clicks "Confirm Registration"
- **Then** The registration service executes atomic seat reservation in Redis Lua
- **And** The transaction is committed to the relational database
- **And** Returns HTTP status 200 OK with message "Successfully enrolled in CSC10002_01"
- **And** The active seat count becomes 39/40 for section "CSC10002_01"
- **And** Both Theory and Practice sessions appear automatically on the student's personal weekly timetable

Scenario AC-F03-02: Registration Rejected When Section Capacity is Exhausted (0 Over-allocations)

Scenario: System prevents enrollment when section capacity is exhausted

- **Given** The student is authenticated and the registration window is "OPEN"
- **And** Section "CSC10002_01" has reached full capacity of 40/40 enrolled students
- **When** The student attempts to enroll in section "CSC10002_01"
- **Then** The Redis Lua script rejects the operation with error "CAPACITY_EXCEEDED"
- **And** The backend returns HTTP 409 Conflict with payload: '{"code": "SECTION_CAPACITY_EXCEEDED", "message": "Selected section CSC10002_01 is full"}'
- **And** Section seat quota is not decremented
- **And** The student's enrolled courses remain unchanged

Scenario AC-F03-03: Timetable Conflict Advisory Warning Displayed (ADR-004)

Scenario: Student selects an overlapping section without blocking enrollment

- **Given** The student is already enrolled in section "MTH00001_01" scheduled on "Tuesday 07:30 - 09:10"
- **And** The student selects section "PHY00002_03" scheduled on "Tuesday 08:20 - 10:00"
- **When** The student views the registration selection or schedule planner
- **Then** The system flags a "Schedule Conflict Advisory" for Tuesday Periods 2-3
- **And** The conflict is visually highlighted in amber with an explanatory warning
- **And** The "Submit Registration" action remains enabled and accessible

Scenario AC-F03-04: Cross-Track Cohort Isolation Enforced (ADR-008)

Scenario: Standard track student attempts to enroll in a Project-Based section

- **Given** The student "24127003" has academic program track "STANDARD"
- **And** Section "CSC10002_CLC01" has target_program_type = "PROJECT_BASED"
- **And** The section has is_cross_enrollment_allowed = FALSE
- **When** The student attempts to submit registration for "CSC10002_CLC01"
- **Then** The registration service rejects the request with HTTP 403 Forbidden
- **And** Returns error message: "Enrollment restricted: Section is reserved for Project-Based students"

Feature F04: Timetable Visualisation & Scheduling (FR-13)

Scenario AC-F04-01: Weekly Timetable Displays Theory and Practice Sessions with Facility Details

Scenario: Student inspects personalized weekly timetable for active term

- **Given** Student "24127003" is enrolled in section "CSC10002_01"
- **And** "CSC10002_01" has Theory scheduled on "Monday 07:30 - 10:00" at "CS2_C32"
- **And** "CSC10002_01" has Practice scheduled on "Wednesday 13:00 - 16:30" at "CS2_Lab04"
- **When** The student navigates to "/student/timetable" and selects week 5
- **Then** The timetable grid displays the Monday lecture block colored by subject
- **And** Displays the Wednesday lab block with distinct lab indicator icon
- **And** Each timetable cell indicates Course Title, Section Code, Room, Campus, and Lecturer Name

Feature F05: Gradebook & Dynamic Evaluation (FR-17, FR-18, FR-19)

Scenario AC-F05-01: Primary Lecturer Submits Dynamic Rubric Grade Sheet (Happy Path)

Scenario: Assigned primary lecturer locks and submits section grades for academic review

- **Given** Lecturer "GV001" is assigned as Primary Lecturer (is_primary = TRUE) to section "CSC10002_01"
- **And** The section grade sheet is in "Draft" state
- **And** Grade components (Attendance [10%], Midterm [30%], Lab [20%], Final [40%]) sum to exactly 100%
- **And** All enrolled students have valid numeric scores populated between 0.0 and 10.0
- **When** Lecturer "GV001" clicks "Submit to Academic Affairs"
- **Then** The grade sheet state transitions from "Draft" to "Submitted"
- **And** Lecturer "GV001" and assigned TAs are restricted to read-only access on the sheet
- **And** An immutable audit log entry is appended with actor "GV001" and timestamp
- **And** A notification is dispatched to the Academic Affairs review queue

Scenario AC-F05-02: Academic Affairs Finalizes Grades and Recalculates Cumulative GPA

Scenario: Academic Affairs approves submitted grade sheet and publishes official grades

- **Given** A grade sheet for section "CSC10002_01" exists in "Submitted" state
- **And** An officer with role "Academic Affairs" is logged into the management portal
- **When** The officer clicks "Approve & Finalize Grade Sheet"
- **Then** The grade sheet state transitions to "Finalized"
- **And** The official grades immediately become visible to all enrolled students via their Grade Portal
- **And** Cumulative GPA and completed credits for affected students are automatically recalculated

Scenario AC-F05-03: Direct Mutation of Finalized Grade Rejected via API (Edge Case)

Scenario: User attempts to edit a finalized grade directly via API

- **Given** Grade for student "24127003" in section "CSC10002_01" is in "Finalized" state
- **When** A PUT request is dispatched to "/api/grades/sections/CSC10002_01/students/24127003"
- **Then** The backend rejects the request with HTTP 403 Forbidden
- **And** Returns error message: "Finalized grades cannot be directly edited. Submit a Grade Rectification Request."

Scenario AC-F05-04: Teaching Assistant Restricted from Submitting Official Grade Sheet

Scenario: Teaching Assistant attempts to finalize or submit grade sheet

- **Given** User "TA001" is assigned to section "CSC10002_01" with is_primary = FALSE
- **When** User "TA001" dispatches a POST request to "/api/grades/sections/CSC10002_01/submit"
- **Then** The backend rejects the request with HTTP 403 Forbidden
- **And** Returns error message: "Only the designated Primary Lecturer can submit section grade sheets"

Feature F06: Student Affairs & Document Workflows (FR-21, FR-22, FR-23)

Scenario AC-F06-01: Student Submits Administrative Document Request

Scenario: Student submits an administrative document request successfully

- **Given** Student "24127003" is authenticated and accesses the Document Request portal
- **When** The student submits a request of type "ENROLLMENT_VERIFICATION" with reason "Military deferment"
- **Then** A new request record is created with a unique tracking ID (e.g., "REQ-2026-0042")
- **And** Initial status is set to "Submitted"
- **And** The request appears in the student's "My Requests" tracking list

- **And** The request appears in the Student Affairs (CTSV) pending queue

Scenario AC-F06-02: Administrative Staff Reviews and Approves Document Request

Scenario: CTSV officer approves student document request and marks ready for pickup

- **Given** A document request "REQ-2026-0042" exists with status "Under_Review"
- **And** A CTSV officer inspects the attached military board call notice
- **When** The officer clicks "Approve Request" and specifies pickup date "2026-11-05"
- **Then** The request status transitions to "Approved"
- **And** An automated in-app notification is dispatched to student "24127003"
- **And** The audit history records approver ID and timestamp

Feature F07: Academic Operations & Term Management (FR-25, FR-26)

Scenario AC-F07-01: Academic Staff Opens Track-Specific Registration Window

Scenario: Phòng Đào tạo opens course registration for Standard Track

- **Given** An officer with role "Academic Affairs" accesses "/admin/registration-windows"
- **When** The officer creates a registration window for semester "2026-1", program "STANDARD", cohort "2024"
- **And** Sets start time to "2026-11-01 08:00" and end time to "2026-11-07 17:00"
- **And** Sets status to "OPEN"
- **Then** The system activates the registration window in registration.registration_windows
- **And** Eligible Standard students are granted access to enroll in sections
- **And** Project-Based students attempting to register under this window are excluded

Feature F08: Asynchronous Notifications & System Alerts (FR-28, FR-29)

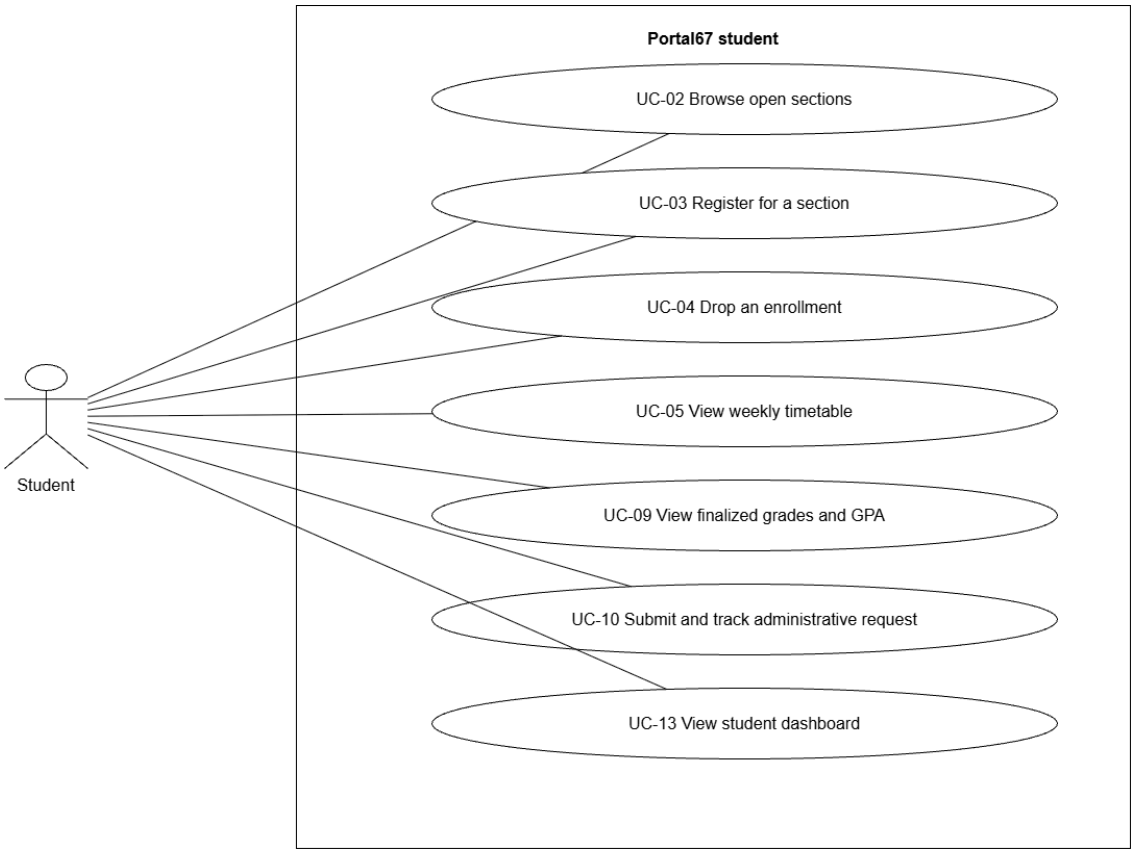
Scenario AC-F08-01: Asynchronous Dispatch and Rendering of Grade Finalization Notification

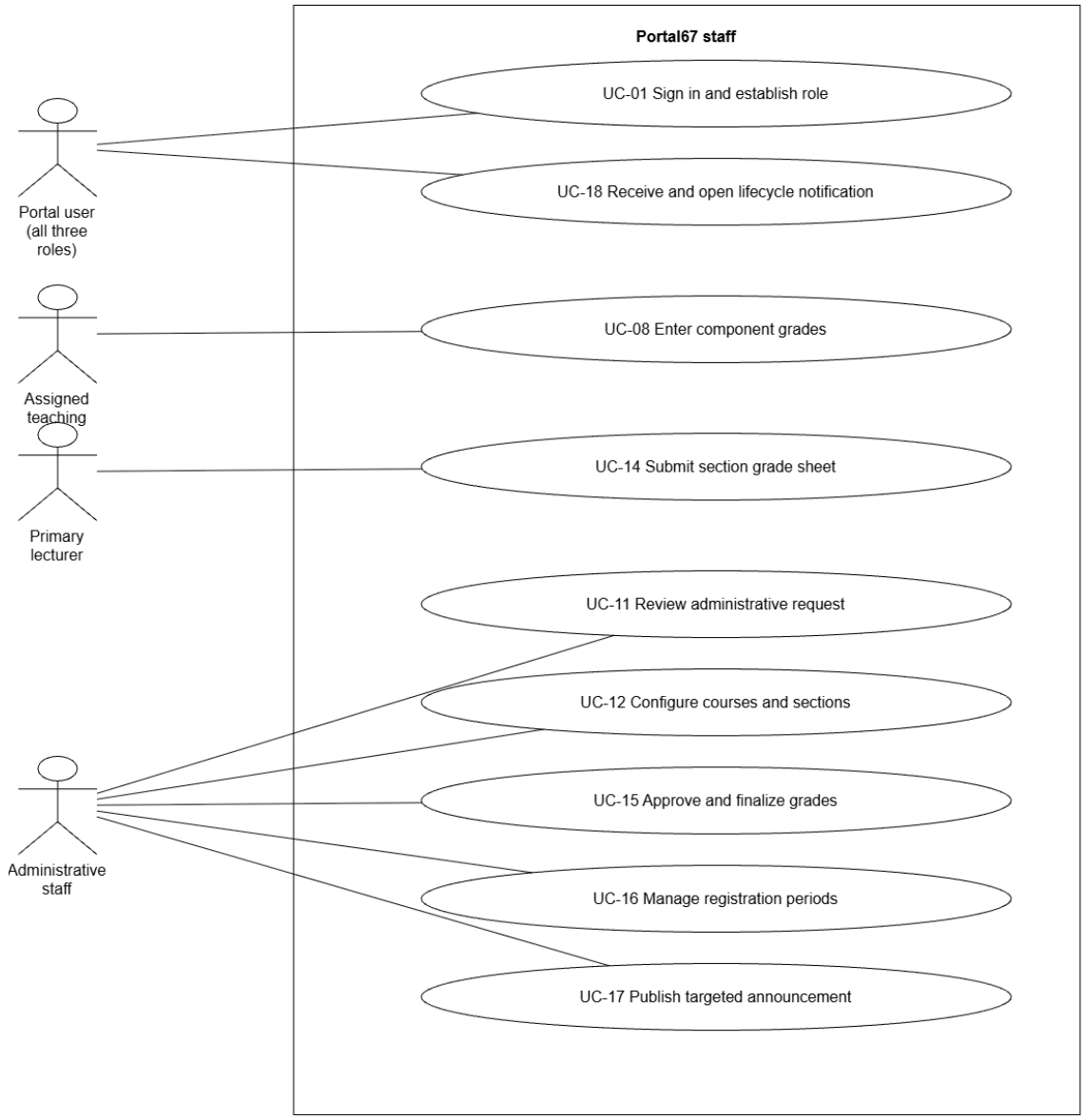
Scenario: System delivers real-time notification to enrolled students when grades are published

- **Given** Section "CSC10002_01" grade sheet transitions to "Finalized"
- **When** The background notification worker consumes the "GradeFinalizedEvent" from Redis Streams
- **Then** An in-app alert is created in affairs.notifications for all enrolled students
- **And** When student "24127003" refreshes or opens the portal
- **And** A bell notification badge appears with message: "Official grades for CSC10002 have been published"
- **And** Clicking the notification routes directly to "/student/grades"

7 Use Case Model

7.1 Use Case Diagram





7.2 Actors

Actor	Description
Student	STANDARD or PROJECT-BASED student, scoped by track and cohort.
Teaching staff	Assigned lecturers, co-teachers and TAs edit Draft component grades.
Primary lecturer	Only the designated primary lecturer submits the grade sheet.

Administrative staff	Authorized academic setup, window, grade finalization, request and announcement operations.
Institutional identity provider	Supports institutional OAuth/SSO.

7.3 Use Case List

ID	Use case	Primary actor	Priority	Related FR
UC-01	Sign in and establish role	Portal user	Must	FR-01, FR-02, FR-03
UC-02	Browse open sections	Student	Must	FR-06, FR-03
UC-03	Register for a section	Student	Must	FR-07, FR-09, FR-03
UC-04	Drop an enrollment	Student	Must	FR-08, FR-03
UC-05	View weekly timetable	Student	Must	FR-13, FR-03
UC-06	View tuition obligations	Student	Should	FR-27, FR-03
UC-07	Update simulated payment status	Administrative staff	Should	FR-27, FR-03
UC-08	Enter component grades	Assigned lecturer or teaching assistant	Must	FR-17, FR-03
UC-09	View finalized grades and GPA	Student	Must	FR-15, FR-16, FR-03
UC-10	Submit and track administrative request	Student	Must	FR-21, FR-23, FR-03
UC-11	Review administrative request	Administrative staff	Must	FR-22, FR-23, FR-28, FR-03
UC-12	Configure courses and sections	Administrative staff	Must	FR-26, FR-03
UC-13	View student dashboard	Student	Must	FR-04, FR-03
UC-14	Submit section grade sheet	Primary lecturer	Must	FR-18, FR-03
UC-15	Approve and finalize grades	Academic Affairs staff	Must	FR-19, FR-28, FR-03
UC-16	Manage registration periods	Academic Affairs staff	Must	FR-25, FR-28, FR-03

UC-17	Publish targeted announcement	Administrative staff	Must	FR-29, FR-03
UC-18	Receive and open lifecycle notification	Portal user	Must	FR-29, FR-03
UC-19	View and update contact profile	Student	Should	FR-05, FR-03
UC-20	Plan trial timetable	Student	Should	FR-10, FR-03
UC-21	Submit cross-registration petition	PROJECT-BASED student	Should	FR-12, FR-03
UC-22	Correct finalized grade through approval	Teaching staff and approving officer	Should	FR-20, FR-03
UC-23	Lodge grade review request	Student	Should	FR-24, FR-03
UC-24	Submit ranked specialization preferences	Qualified Year 2 student	Could	FR-11, FR-03
UC-25	Export personal timetable	Student	Could	FR-14, FR-03

8 Use Case Specifications

8.1 UC-01 Sign in and establish role

Use case ID	UC-01
Use case name	Sign in and establish role
Brief description	Sign in and establish role.
Primary actor	Portal user
Secondary actors	Institutional identity provider (OAuth alternative).
Priority	Must
Related requirements	FR-01, FR-02, FR-03
Trigger	The actor starts main-flow step 1.
Pre-conditions	A pre-provisioned active account exists.
Post-conditions	A valid access token (at most 15 minutes) and refresh token (at most 7 days) identify the user and role.
Main flow	1. User enters institutional identifier and password. 2. Portal67 validates credentials and account status. 3. Portal67 resolves role and student track/cohort where applicable. 4. Portal67 issues scoped tokens and opens the role dashboard.
Alternative flows	**A1 (step 1): Institutional OAuth/SSO:** 1. User authenticates with the institutional provider. 2. Portal67 verifies the callback and maps it to a pre-provisioned account. 3. Continue at main-flow step 3. **A2 (step 1): Demo switcher (Should, FR-02):** 1. Evaluator selects a seeded role. 2. Portal67 establishes that demo identity and continues at main-flow step 3.
Exception flows	**E1 (step 2): Invalid credentials or inactive account:** 1. Portal67 denies sign-in without issuing tokens. 2. User may retry step 1. **E2 (step 3): Identity is not provisioned:** 1. Portal67 denies access. 2. The use case ends; no public self-registration is offered.
Business rules	BR-18, BR-20
Non-functional constraints	NFR-05, NFR-06, NFR-07

8.2 UC-02 Browse open sections

Use case ID	**UC-02**
Use case name	Browse open sections
Brief description	Browse open sections.
Primary actor	Student
Secondary actors	None.
Priority	Must
Related requirements	FR-06, FR-03
Trigger	The actor starts main-flow step 1.
Pre-conditions	Student is signed in.
Post-conditions	Eligible offerings and integrated session schedules are displayed.
Main flow	1. Student opens offerings and selects term. 2. Portal67 loads the catalog partition for the student track and cohort. 3. Student searches or filters by department, campus and course. 4. Portal67 shows sections, capacity and integrated Theory-Practice times.
Alternative flows	**A1 (step 3): Change search filters:** 1. Student changes filters. 2. Portal67 refreshes results and continues at step 4.
Exception flows	**E1 (step 4): No matching sections:** 1. Portal67 shows an empty state. 2. Student may return to step 3. **E2 (step 2): Catalog cannot be loaded:** 1. Portal67 displays a retryable error. 2. Student retries step 2 or exits.
Business rules	BR-06, BR-07
Non-functional constraints	NFR-01, NFR-07, NFR-08

8.3 UC-03 Register for a section

Use case ID	**UC-03**
Use case name	Register for a section
Brief description	Register for a section.
Primary actor	Student
Secondary actors	None.

Use case ID	**UC-03**
Priority	Must
Related requirements	FR-07, FR-09, FR-03
Trigger	The actor starts main-flow step 1.
Pre-conditions	An eligible track/cohort registration window is OPEN.
Post-conditions	One section seat and all required Theory-Practice sessions are enrolled; no over-allocation occurs.
Main flow	1. Student selects a section. 2. Portal67 validates the OPEN window, track/cohort eligibility, duplicate root course and available capacity. 3. Portal67 checks all Theory-Practice meetings for overlaps and displays any CONFLICT_DETECTED amber advisory. 4. Student confirms registration; an overlap warning leaves submission enabled. 5. Portal67 reserves one section seat atomically through Redis Lua and initiates relational persistence. 6. Portal67 confirms the committed enrollment and updates all integrated timetable sessions.
Alternative flows	**A1 (step 3): Selected section overlaps an enrollment:** 1. Portal67 highlights the overlap and explains the advisory. 2. Student can proceed at main-flow step 4 or choose a different section at step 1.
Exception flows	**E1 (step 2): Window, duplicate-course, track or capacity rule fails:** 1. Portal67 rejects the request with the specific reason. 2. No enrollment or seat allocation changes. **E2 (step 5): Final seat is taken or persistence fails:** 1. Portal67 rejects a lost seat race or compensates the failed operation. 2. Portal67 avoids partial enrollment and displays the current outcome. 3. Retry starts at step 1 after reconciliation.
Business rules	BR-01, BR-02, BR-03, BR-04, BR-06, BR-07, BR-08
Non-functional constraints	NFR-02, NFR-03, NFR-04, NFR-11

8.4 UC-04 Drop an enrollment

Use case ID	**UC-04**
Use case name	Drop an enrollment
Brief description	Drop an enrollment.

Use case ID	UC-04
Primary actor	Student
Secondary actors	None.
Priority	Must
Related requirements	FR-08, FR-03
Trigger	The actor starts main-flow step 1.
Pre-conditions	Student owns the enrollment and the applicable registration/drop period is OPEN.
Post-conditions	The section seat and both Theory-Practice sessions are released consistently.
Main flow	1. Student opens current enrollments. 2. Student selects a section and confirms Drop. 3. Portal67 checks ownership and the permitted track/cohort window. 4. Portal67 atomically releases the section seat and updates relational enrollment records. 5. Portal67 removes all integrated sessions from the student timetable and confirms.
Alternative flows	**A1 (step 2): Cancel dropping:** 1. Student cancels. 2. Portal67 retains the enrollment; the use case ends.
Exception flows	**E1 (step 3): Drop is not permitted:** 1. Portal67 displays the failed ownership/window rule. 2. Enrollment remains unchanged. **E2 (step 4): Drop persistence fails:** 1. Portal67 compensates or reconciles the release without a partial drop. 2. Portal67 shows a retryable failure and the authoritative enrollment state.
Business rules	BR-01, BR-07, BR-08
Non-functional constraints	NFR-04, NFR-11

8.5 UC-05 View weekly timetable

Use case ID	UC-05
Use case name	View weekly timetable
Brief description	View weekly timetable.
Primary actor	Student
Secondary actors	None.
Priority	Must

Use case ID	**UC-05**
Related requirements	FR-13, FR-03
Trigger	The actor starts main-flow step 1.
Pre-conditions	Student is signed in.
Post-conditions	The selected weekly timetable displays all enrolled Theory-Practice sessions.
Main flow	1. Student chooses semester, week and study day. 2. Portal67 loads the student enrolled section sessions. 3. Portal67 displays course, section, room, building, campus CS1/CS2 and instructor. 4. Portal67 highlights overlapping meetings as advisory information.
Alternative flows	**A1 (step 1): Change timetable filters:** 1. Student changes week or day. 2. Continue at main-flow step 2.
Exception flows	**E1 (step 2): No sessions exist:** 1. Portal67 displays an empty timetable. 2. Student may change filters at step 1. **E2 (step 2): Timetable retrieval fails:** 1. Portal67 displays a retryable error. 2. Student retries step 2 or exits.
Business rules	BR-03, BR-07
Non-functional constraints	NFR-08, NFR-10

8.6 UC-08 Enter component grades

Use case ID	**UC-08**
Use case name	Enter component grades
Brief description	Enter component grades.
Primary actor	Assigned lecturer or teaching assistant
Secondary actors	None.
Priority	Must
Related requirements	FR-17, FR-03
Trigger	The actor starts main-flow step 1.
Pre-conditions	The user is assigned to the section and its grade sheet is Draft.

Use case ID	UC-08
Post-conditions	Valid component scores are stored without publishing official grades.
Main flow	1. Teaching staff selects an assigned section. 2. Portal67 loads enrolled students and dynamic assessment components. 3. Teaching staff enters component scores. 4. Portal67 checks assignment, Draft status and scores from 0.0 to 10.0 with one decimal place. 5. Portal67 saves valid scores and recalculates draft course results.
Alternative flows	**A1 (step 3): Edit selected students:** 1. Teaching staff changes selected scores. 2. Other students remain unchanged; continue at step 4.
Exception flows	**E1 (step 4): Invalid score, unauthorized user or non-Draft sheet:** 1. Portal67 rejects the edit and explains the reason. 2. No prohibited change is stored. **E2 (step 5): Save fails:** 1. Portal67 preserves the previous grades. 2. Teaching staff may retry from step 3.
Business rules	BR-09, BR-10, BR-12, BR-13, BR-14
Non-functional constraints	NFR-07, NFR-11

8.7 UC-09 View finalized grades and GPA

Use case ID	UC-09
Use case name	View finalized grades and GPA
Brief description	View finalized grades and GPA.
Primary actor	Student
Secondary actors	None.
Priority	Must
Related requirements	FR-15, FR-16, FR-03
Trigger	The actor starts main-flow step 1.
Pre-conditions	Student is signed in.
Post-conditions	Published grades, credit totals and GPA are displayed; hypothetical values never change official grades.

Use case ID	UC-09
Main flow	1. Student selects an academic term. 2. Portal67 retrieves the student finalized course grades and dynamic component breakdown. 3. Portal67 calculates semester and cumulative credit-weighted GPA to two decimals, using the highest grade for retaken courses. 4. Portal67 displays 10-point scores, letter grades, 4-point GPA and accumulated credits.
Alternative flows	**A1 (step 4): What-if GPA (Should, FR-16):** 1. Student enters hypothetical target scores for the current term. 2. Portal67 calculates a clearly labeled forecast without modifying official results.
Exception flows	**E1 (step 2): Term is not finalized or has no results:** 1. Portal67 explains that official grades are unavailable. 2. No draft grades are presented as finalized. **E2 (step 3): Grade conversion policy is incomplete:** 1. Portal67 displays available finalized data and identifies the missing calculation rule. 2. Portal67 withholds unsupported GPA values.
Business rules	BR-09, BR-11, BR-12, BR-13
Non-functional constraints	NFR-01, NFR-07

8.8 UC-10 Submit and track administrative request

Use case ID	UC-10
Use case name	Submit and track administrative request
Brief description	Submit and track administrative request.
Primary actor	Student
Secondary actors	None.
Priority	Must
Related requirements	FR-21, FR-23, FR-03
Trigger	The actor starts main-flow step 1.
Pre-conditions	Student is signed in and a supported request type is available.
Post-conditions	A Submitted request and immutable history exist; the owner can track its progression.

Use case ID	UC-10
Main flow	1. Student selects verification, military postponement, transcript or tuition extension. 2. Portal67 displays the form and required digital uploads. 3. Student completes fields, attaches documents and submits. 4. Portal67 validates the submission and saves a unique request with Submitted status and audit history. 5. Portal67 displays tracking ID, current status and history.
Alternative flows	**A1 (step 5): Track an existing request:** 1. Student opens My Requests and selects an owned request. 2. Portal67 displays status, history and staff feedback. **A2 (step 5): Respond to Requires_Information:** 1. Student opens the editable request and supplies clarification or attachments. 2. Portal67 preserves prior history and submits the revision using the agreed revision transition.
Exception flows	**E1 (step 4): Submission or upload is invalid:** 1. Portal67 displays field/upload validation errors. 2. Student corrects the form at step 3. **E2 (step 4): Save fails:** 1. Portal67 creates no partial request. 2. Student can retry after the failure is resolved.
Business rules	BR-15, BR-16
Non-functional constraints	NFR-07, NFR-08, NFR-10

8.9 UC-11 Review administrative request

Use case ID	UC-11
Use case name	Review administrative request
Brief description	Review administrative request.
Primary actor	Administrative staff
Secondary actors	None.
Priority	Must
Related requirements	FR-22, FR-23, FR-28, FR-03
Trigger	The actor starts main-flow step 1.
Pre-conditions	Staff has request-review permission and the request is in a permitted review state.
Post-conditions	A permitted decision and audit history are stored and a notification is queued.

Use case ID	UC-11
Main flow	1. Staff filters the central queue and selects a Submitted request. 2. Portal67 records the transition to Under_Review and displays attachments and history. 3. Staff inspects the request and chooses Approve or Reject with relevant decision details. 4. Portal67 validates authorization and the current request state. 5. Portal67 saves Approved or Rejected with actor/time and queues an asynchronous status notification. 6. Portal67 displays the decision to staff and the owner.
Alternative flows	**A1 (step 3): Return for clarification:** 1. Staff records the missing information. 2. Portal67 sets Requires_Information, preserves history and notifies the student. 3. The student continues through UC-10 clarification flow. **A2 (step 6): Complete fulfillment:** 1. Authorized staff records fulfillment of an Approved request. 2. Portal67 records Completed and queues a status notification.
Exception flows	**E1 (step 4): Request state changed concurrently:** 1. Portal67 rejects the stale decision and reloads the record. 2. Staff reviews the current state before deciding again. **E2 (step 5): Decision save fails:** 1. Portal67 preserves the prior state and audit consistency. 2. Staff receives a retryable error.
Business rules	BR-15, BR-16
Non-functional constraints	NFR-07, NFR-11

8.10 UC-12 Configure courses and sections

Use case ID	UC-12
Use case name	Configure courses and sections
Brief description	Configure courses and sections.
Primary actor	Administrative staff
Secondary actors	None.
Priority	Must
Related requirements	FR-26, FR-03
Trigger	The actor starts main-flow step 1.

Use case ID	**UC-12**
Pre-conditions	Staff has academic setup permissions.
Post-conditions	Courses, section sessions, facilities and teaching assignments are valid and saved.
Main flow	1. Staff selects the course and academic term. 2. Staff configures track/cohort, capacity, campus facilities and Theory-Practice sessions. 3. Staff assigns primary lecturer, co-teachers and teaching assistants. 4. Portal67 validates references, required sessions, capacity and assignments. 5. Staff confirms and Portal67 saves the configuration.
Alternative flows	**A1 (step 2): Course has no lab component:** 1. Staff provides the required Theory sessions only. 2. Continue at step 3.
Exception flows	**E1 (step 4): Setup is incomplete or invalid:** 1. Portal67 identifies invalid facilities, schedules or teaching assignments. 2. Staff corrects the configuration at step 2. **E2 (step 5): Save fails:** 1. Portal67 preserves the prior configuration. 2. Staff receives a retryable error.
Business rules	BR-06, BR-07, BR-14
Non-functional constraints	NFR-07, NFR-08

8.11 UC-13 View student dashboard

Use case ID	**UC-13**
Use case name	View student dashboard
Brief description	View student dashboard.
Primary actor	Student
Secondary actors	None.
Priority	Must
Related requirements	FR-04, FR-03
Trigger	The actor starts main-flow step 1.
Pre-conditions	Student is signed in.
Post-conditions	Daily classes, cumulative GPA, earned credits, pending requests and active announcements are displayed.

Use case ID	UC-13
Main flow	1. Student opens the dashboard. 2. Portal67 retrieves the student daily schedule, finalized academic totals and pending requests. 3. Portal67 retrieves announcements visible to the student audience. 4. Portal67 renders the consolidated dashboard and links to detailed views.
Alternative flows	**A1 (step 4): Open a dashboard item:** 1. Student selects a class, grade, request or announcement. 2. Portal67 opens the corresponding authorized view.
Exception flows	**E1 (step 2): A dashboard data source fails:** 1. Portal67 labels the affected information unavailable. 2. Student can retry without seeing fabricated totals.
Business rules	BR-19
Non-functional constraints	NFR-07, NFR-08, NFR-09, NFR-10

8.12 UC-14 Submit section grade sheet

Use case ID	UC-14
Use case name	Submit section grade sheet
Brief description	Submit section grade sheet.
Primary actor	Primary lecturer
Secondary actors	None.
Priority	Must
Related requirements	FR-18, FR-03
Trigger	The actor starts main-flow step 1.
Pre-conditions	The designated primary lecturer owns a Draft sheet with all required scores entered.
Post-conditions	The sheet is Submitted and read-only to assigned lecturers and TAs; the review queue is notified.
Main flow	1. Primary lecturer opens the completed Draft sheet. 2. Portal67 displays components, weights and score completeness. 3. Primary lecturer selects Submit to Academic Affairs. 4. Portal67 checks primary assignment, valid scores and active weights totaling exactly 100%. 5. Portal67 atomically locks the sheet as Submitted, appends audit history and queues a review notification.

Use case ID	UC-14
Alternative flows	**A1 (step 3): Cancel submission:** 1. Lecturer returns to Draft editing. 2. The sheet remains Draft.
Exception flows	**E1 (step 4): TA/co-teacher attempts submission or data are incomplete:** 1. Portal67 denies unauthorized submission or identifies invalid scores/weights. 2. The sheet remains Draft. **E2 (step 5): Submission save fails:** 1. Portal67 preserves the Draft sheet without a partial lock. 2. Lecturer receives a retryable error.
Business rules	BR-09, BR-10, BR-12, BR-14
Non-functional constraints	NFR-07, NFR-11

8.13 UC-15 Approve and finalize grades

Use case ID	UC-15
Use case name	Approve and finalize grades
Brief description	Approve and finalize grades.
Primary actor	Academic Affairs staff
Secondary actors	None.
Priority	Must
Related requirements	FR-19, FR-28, FR-03
Trigger	The actor starts main-flow step 1.
Pre-conditions	An authorized officer selects a Submitted sheet.
Post-conditions	The sheet is Finalized, official grades are visible, GPA/credits update and student notifications are queued.
Main flow	1. Staff opens the submitted grade-sheet queue. 2. Staff inspects a selected sheet and its components. 3. Staff selects Approve and Finalize. 4. Portal67 validates staff authority and Submitted status. 5. Portal67 finalizes the sheet, recalculates affected GPA and credits and stores audit history consistently. 6. Portal67 publishes the results and queues grade-finalized notifications.

Use case ID	UC-15
Alternative flows	**A1 (step 3): Defer the decision:** 1. Staff exits without approval. 2. The sheet remains Submitted.
Exception flows	**E1 (step 4): Invalid role or stale sheet status:** 1. Portal67 denies finalization and shows the current state. 2. No grades change. **E2 (step 5): Finalization or recalculation fails:** 1. Portal67 avoids partial publication and preserves the previous consistent state. 2. Staff receives a retryable error.
Business rules	BR-11, BR-12, BR-13
Non-functional constraints	NFR-07, NFR-11

8.14 UC-16 Manage registration periods

Use case ID	UC-16
Use case name	Manage registration periods
Brief description	Manage registration periods.
Primary actor	Academic Affairs staff
Secondary actors	None.
Priority	Must
Related requirements	FR-25, FR-28, FR-03
Trigger	The actor starts main-flow step 1.
Pre-conditions	Staff has term-management permission.
Post-conditions	Window dates, track/cohort scope and operational state are saved and enforced.
Main flow	1. Staff selects semester, track and cohort. 2. Staff configures dates and chooses Open, Suspend or Close. 3. Portal67 validates dates, audience and administrative permission. 4. Portal67 saves the window state and enforces it for registration and dropping. 5. For an opened window, Portal67 queues notifications for eligible students.

Use case ID	UC-16
Alternative flows	**A1 (step 2): Suspend or close an existing window:** 1. Staff selects Suspend or Close. 2. Portal67 prevents further operations requiring an OPEN window after step 4.
Exception flows	**E1 (step 3): Window configuration is invalid:** 1. Portal67 explains the date or audience error. 2. Staff corrects the configuration at step 2. **E2 (step 4): Window update fails:** 1. Portal67 retains the previous state. 2. Staff receives a retryable error.
Business rules	BR-01, BR-06
Non-functional constraints	NFR-07, NFR-11

8.15 UC-17 Publish targeted announcement

Use case ID	UC-17
Use case name	Publish targeted announcement
Brief description	Publish targeted announcement.
Primary actor	Administrative staff
Secondary actors	None.
Priority	Must
Related requirements	FR-29, FR-03
Trigger	The actor starts main-flow step 1.
Pre-conditions	Staff has announcement permissions.
Post-conditions	A markdown announcement is visible only to its intended audience.
Main flow	1. Staff opens announcement authoring. 2. Staff enters markdown content and selects global, program-specific or faculty-wide audience. 3. Portal67 validates content and audience permissions. 4. Staff confirms publication. 5. Portal67 saves the announcement and displays it on authorized dashboards.
Alternative flows	**A1 (step 4): Cancel publication:** 1. Staff cancels. 2. No announcement is published.

Use case ID	UC-17
Exception flows	**E1 (step 3): Content or audience is invalid:** 1. Portal67 identifies the error. 2. Staff corrects the announcement at step 2. **E2 (step 5): Publication fails:** 1. Portal67 avoids partial publication. 2. Staff receives a retryable error.
Business rules	None
Non-functional constraints	NFR-07, NFR-08, NFR-10

8.16 UC-18 Receive and open lifecycle notification

Use case ID	UC-18
Use case name	Receive and open lifecycle notification
Brief description	Receive and open lifecycle notification.
Primary actor	Portal user
Secondary actors	None.
Priority	Must
Related requirements	FR-28, FR-03
Trigger	The actor starts main-flow step 1; UC-18 is triggered by a committed lifecycle event.
Pre-conditions	A grade-finalized, request-status or registration-open event has been committed.
Post-conditions	An authorized recipient can see the in-app alert and open its related record.
Main flow	1. Portal67 queues the committed lifecycle event for background processing. 2. A background worker resolves authorized recipients and creates in-app notifications. 3. User opens or refreshes the portal and sees the notification badge. 4. User selects a notification. 5. Portal67 checks record access and opens the related grades, request or registration view.
Alternative flows	**A1 (step 3): No new notification:** 1. Portal67 shows the current notification list. 2. The user continues normal portal use.

Use case ID	UC-18
Exception flows	**E1 (step 2): Worker delivery fails:** 1. Portal67 retains the committed business result. 2. Delivery is retried under the agreed queue policy; no false business rollback is reported. **E2 (step 5): Target record is no longer accessible:** 1. Portal67 denies record access without exposing private data. 2. User returns to the notification list.
Business rules	BR-12, BR-15
Non-functional constraints	NFR-07, NFR-15

9 Prototype / Mockups

Screen	Related use case(s)	Mockup / link
UI-01 - Sign in (all roles)	UC-01	Open mockup - Sign-in form, demo-account dropdown and error state.
UI-02 - Today (student)	UC-13, UC-18	Open mockup - Daily overview with notification and account popovers.
UI-03 - Registration (student)	UC-02, UC-03, UC-04, UC-20, UC-21	Open mockup - Course/registration/planner tabs; selection, review, drop and petition dialogs.
UI-04 - Schedule (student)	UC-05	Open mockup - Semester timetable, meeting details and class popovers.
UI-05 - Exams (student)	FR-13; UC-10 for postponement	Open mockup - Midterm/final/deferred tabs, exam details and postponement dialog.
UI-06 - Grades (student)	UC-09, UC-23	Open mockup - Semester summary/detail/forecast tabs; grade-scale and review-request dialogs.

UI-07 - Requests (student)	UC-10	Open mockup - History, request-type forms and status drawers.
UI-08 - Tuition (student)	UC-06, UC-10	Open mockup - Charges, payments, balance and extension-request dialog.
UI-09 - My profile (student)	UC-19	Open mockup - Detailed profile with contact, correction and custom-field dialogs.
UI-10 - My teaching (lecturer/TA)	UC-08, UC-14	Open mockup - Assigned sections and links to grade entry.
UI-11 - Grade entry (lecturer/TA)	UC-08, UC-14	Open mockup - Component scores, draft/error states and submission confirmation.
UI-12 - Work queue (staff)	UC-11, UC-15, UC-16, UC-18	Open mockup - Pending work and notification shortcuts.
UI-13 - Student requests (staff)	UC-11	Open mockup - Review drawers, clarification/rejection dialogs and fulfillment states.
UI-14 - Grade approvals (staff)	UC-15	Open mockup - Review queue/detail and return/finalize confirmations.
UI-15 - Courses and sections (staff)	UC-12	Open mockup - Section list and theory/practice editing dialogs.
UI-16 - Registration periods (staff)	UC-16	Open mockup - Window configuration and Open/Suspend confirmations.
UI-17 - Announcements (staff)	UC-17	Open mockup - Announcement list, composition, preview and publish dialogs.
UI-18 - Tuition records (staff)	UC-07	Open mockup - Payment lookup, entry and confirmation dialogs.

10 Requirements Traceability Matrix

[This matrix is extended in PA3 (design elements) and PA4 (test cases).]

Req. ID	Use case	Screen	Design element (PA3)	Test cases (PA4)
FR-01	UC-01	UI-01	To be defined in PA3	AC-F01-01, AC-F01-03
FR-02	UC-01	UI-01	To be defined in PA3	To be defined in PA4
FR-03	UC-01 (cross-cutting)	All protected screens	To be defined in PA3	AC-F01-02
FR-04	UC-13	UI-02	To be defined in PA3	AC-F02-01
FR-05	UC-19	UI-09	To be defined in PA3	To be defined in PA4
FR-06	UC-02	UI-03	To be defined in PA3	To be defined in PA4
FR-07	UC-03	UI-03	To be defined in PA3	AC-F03-01, AC-F03-02, AC-F03-04
FR-08	UC-04	UI-03	To be defined in PA3	To be defined in PA4
FR-09	UC-03	UI-03	To be defined in PA3	AC-F03-03
FR-10	UC-20	UI-03	To be defined in PA3	To be defined in PA4
FR-11	UC-24	Not specified	To be defined in PA3	To be defined in PA4
FR-12	UC-21	UI-03	To be defined in PA3	To be defined in PA4
FR-13	UC-05	UI-04	To be defined in PA3	AC-F04-01
FR-14	UC-25	UI-04	To be defined in PA3	To be defined in PA4
FR-15	UC-09	UI-06	To be defined in PA3	To be defined in PA4
FR-16	UC-09	UI-06	To be defined in PA3	To be defined in PA4
FR-17	UC-08	UI-10, UI-11	To be defined in PA3	AC-F05-01, AC-F05-04
FR-18	UC-14	UI-10, UI-11	To be defined in PA3	AC-F05-01, AC-F05-04
FR-19	UC-15	UI-12, UI-14	To be defined in PA3	AC-F05-02
FR-20	UC-22	UI-14	To be defined in PA3	AC-F05-03*
FR-21	UC-10	UI-07, UI-08	To be defined in PA3	AC-F06-01
FR-22	UC-11	UI-12, UI-13	To be defined in PA3	AC-F06-02

FR-23	UC-10, UC-11	UI-07, UI-13	To be defined in PA3	AC-F06-01, AC-F06-02
FR-24	UC-23	UI-06	To be defined in PA3	To be defined in PA4
FR-25	UC-16	UI-12, UI-16	To be defined in PA3	AC-F07-01
FR-26	UC-12	UI-15	To be defined in PA3	To be defined in PA4
FR-27	UC-06, UC-07	UI-08, UI-18	To be defined in PA3	To be defined in PA4
FR-28	UC-18	UI-02, UI-12	To be defined in PA3	AC-F08-01
FR-29	UC-17	UI-17, UI-02	To be defined in PA3	To be defined in PA4

11 Requirements Review

#	Issue found	Requirement(s)	Resolution
1	Nine use cases are listed in the RTM but have no specification.	UC-06, 07, 19-25	Write the missing specs
2	NFR-01 and NFR-03 have unclear units. NFR-02 mixes two metrics.	NFR-01, 02, 03	Specify rate and duration, and split NFR-02.
3	FR-11 has no screen.	FR-11	Add a mockup.
4	UI-05 is an orphaned screen with no reference yet	UI-05	Add reference to screen
5	5 “Must” FRs have no acceptance criteria, and only 15 of 29 FRs have any test.	FR-06, 08, 15, 26, 29	Add scenarios and remap tests in the RTM
6	Business rule contradiction: “100% ACID” conflicts with asynchronous write-behind persistence	NFR-11, BR-08, UC-03	Reword to “no partial or lost enrollments” and define when confirmation is shown.

12 AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Đoàn Quốc Bảo	Gemini 3.8 Flash	Explain many Use case specification	
2	Lê Minh Đức	ChatGPT	For drafting & fixing Problem statement, Stakeholders.	Comparing with PA1, manual reading and checking.
3	Đoàn Quốc Bảo	Claude Sonnet 5.5	Reviewed the full SRS and cross-checked the RTM against the use case list, specs, mockups and acceptance criteria.	Checked each finding against the document for missing specs, incorrect references, ..